Does Cross-Attention Fusion of ECG and Pulse Signals Verify ICU Arrhythmia Alarms? A Leakage-Audited Study on PhysioNet/CinC 2015 and VTaC

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***Abstract*—ICU monitors raise many false arrhythmia alarms, and alarm fatigue can delay the response to the true ones. We ask whether a second, independent pulse signal (the photoplethysmogram, PPG) can verify an alarm triggered by the ECG, and whether cross-attention is the right way to combine the two signals. We compared 14 neural architectures, six classical algorithms on 266 engineered features, a hybrid attention CNN-GRU recipe and several ensembles on two open datasets: PhysioNet/CinC Challenge 2015 (a 592-record PPG cohort) and VTaC (4,542 ventricular tachycardia alarms). The protocol used inner validation sets, 10 seeds and corrected statistics. Fixing a validation leak in our first code version lowered the F1 of the proposed cross-attention model from 0.774 to 0.699 and removed its advantage over concatenation (ΔF1 $= +0.006$, 95% CI -0.015 to $+0.028$). Classifiers on engineered features (F1 up to 0.818) and a hybrid deep recipe (F1 0.828 on Challenge 2015; F1 0.858 and AUC 0.966 on VTaC) clearly beat plain cross-attention. Counterfactual tests show that the fused models ignore ECG-to-pulse timing: swapping in another patient's same-label PPG changes AUC by less than 0.03, and attention does not follow the measured pulse transit time ($\rho = -0.04$). Transfer between datasets is lopsided. Models trained on Challenge 2015 reach an F1 of only 0.56 to 0.60 on VTaC when they use the ECG. In these data the pulse acts as a patient-agnostic evidence cue, not as a timing check. Training and feature engineering matter more than the fusion layer, and results from a single dataset overstate generalization. This is a research prototype without prospective validation.**

***Index Terms*—false alarm reduction, intensive care unit, arrhythmia alarm verification, multimodal fusion, cross-attention, electrocardiogram, photoplethysmogram, data leakage, pulse transit time, PhysioNet**

I. Introduction

Bedside monitors in intensive care units (ICUs) raise many arrhythmia alarms, and a large share of them are false. An observational study of consecutive ICU patients and a review of the alarm-fatigue literature both report that this burden desensitizes staff and can slow the response to true alarms [1], [2]. Ventricular tachycardia (VT) alarms are among the most frequent life-threatening ones, and false VT alarms have been the hardest to suppress reliably [3], [4]. Most monitors decide from the electrocardiogram (ECG) alone. Electrode motion or muscle noise corrupts the ECG, and it corrupts the very signal that would be used to double-check the alarm.

A second, physically independent signal can help. The photoplethysmogram (PPG) and the arterial blood pressure (ABP) waveform show whether an electrical event was followed by a mechanical pulse, and they fail under different conditions than the ECG. A genuine arrhythmia should disturb both signals, while an ECG artifact should leave the pulse intact. Earlier work used this idea with hand-crafted ABP features [5], in the 2015 PhysioNet/Computing in Cardiology Challenge [4], [6], and with deep models. The closest deep models are the attention-based convolutional and recurrent networks of Mousavi et al. [7], which reported 93.88% sensitivity and 92.05% specificity with three signals, and the contrastive approach of Zhou et al. [8], which adds learned alarm-type embeddings.

We began with a specific hypothesis. A transformer-style *cross-attention* layer [9], in which ECG features query PPG features, should learn the pulse transit time (PTT) between the electrical beat and the arriving pulse. Comparing the two signals this way should verify alarms better than a model that sees one signal or simply concatenates both. This gives two research questions: (i) does fusing ECG and pulse improve alarm verification, and (ii) if so, does the attention-based alignment explain the gain? Our first code version answered (ii) with a yes (F1 of 0.774 for cross-attention), but that number was biased upward, because the best epoch was selected on the same fold that was reported. This paper starts from a re-run under a leakage-free protocol.

We set five objectives: (1) compare cross-attention with ECG-only, PPG-only and concatenation baselines at matched parameter count; (2) benchmark it against stronger classical, deep, hybrid and ensemble alternatives on two datasets; (3) test with counterfactual perturbations whether fusion models use ECG-to-pulse timing; (4) measure robustness to noise and sensor loss, and transfer across datasets; and (5) report

every result with corrected statistics, including the ones that contradict our hypothesis.

Section II covers the data, preprocessing, models, training, implementation and mathematics. Section III gives the results, discussion and limitations, and Section IV concludes.

II. Methods

A. Overview

Fig. 1 shows the pipeline. Each alarm event becomes a 10 s window of ECG and a pulse channel that ends exactly at the alarm onset. Three families of models are trained on the same windows and splits: neural networks on the raw signals, classical learners on engineered features, and a hybrid deep recipe. Ensembles combine their predictions.

B. Data collection

We collected no new recordings. All data are public and de-identified, and we obtained them from their original hosts. Table I lists every dataset with its citation and source URL. The cohort statistics below come from the audit files that our preprocessing scripts write.

Challenge 2015. The training set has 750 five-minute ICU records at 250 Hz. The alarm fires at the 300 s mark, and an expert labeled each alarm true or false for one of five types: asystole, extreme bradycardia, extreme tachycardia, ventricular tachycardia, and ventricular flutter/fibrillation [4], [6]. The 500-record test set was never released, so record-wise cross-validation is the appropriate evaluation. Of the 750 records, 627 carry PPG, 343 carry ABP and all have a second ECG lead. The original *PPG cohort* has 592 records (37.7% true alarms). The *recovered cohort* has 724 records (39.6% true). It uses ABP as the pulse channel where PPG is missing (132 records) and drops 26 records whose window is too short or nearly flat-lined.

VTaC. VTaC holds 5,037 annotated VT alarm events, each labeled by at least two independent experts [3]; v1.1 also releases the per-annotator votes. The host throttles each connection, so we downloaded only the window $[-60\ \text{s}, +30\ \text{s}]$ around each alarm onset with parallel HTTP range requests (about 1 GB instead of 18 GB). We then excluded 292 events without a pulse channel and 203 low-quality windows, which left 4,542 events from 2,138 patients (28.0% true alarms). The official patient-disjoint split gives 3,662 training, 453 validation and 427 test events. Hospital and device identifiers are not released.

Auxiliary data. MIMIC PERform AF (35 subjects, 125 Hz) serves only an AF-screening task, with 60 evenly spaced 10 s windows per subject (2,100 windows) [12], [13]. The MIT-BIH Noise Stress Test provides real noise for the stress tests [14].

C. Data preparation

Windowing. The headline task window is the 10 s that end exactly at the alarm onset, and no post-alarm samples enter any model input. Only a separate time-to-verdict analysis reads the 30 s after the onset. This decision-time alignment replaced an earlier center-aligned window.

Preprocessing. We resample signals to 250 Hz, apply a zero-phase fourth-order Butterworth band-pass (ECG 0.5 to 40 Hz; pulse 0.5 to 8 Hz), and z-normalize each window. Nothing is fitted across windows, so no train/test statistics leak. A simple per-channel quality score (flat-line variance and clipping fraction) removes windows below a threshold of 0.05.

Splits. Challenge 2015 uses record-wise 5-fold cross-validation repeated for 10 seeds; the seed changes the fold assignment. VTaC uses the official patient-disjoint split with 5 to 10 seeds. Inside every training fold, 15% of the records form an *inner validation set*. It drives early stopping, model selection, calibration, thresholds and ensemble weights, and the outer test fold is evaluated once. All models see identical splits, so comparisons are paired.

D. Model selection

Parameter-matched baseline ladder. The core comparison has four variants matched to about 261k parameters, with the single-signal models widened to compensate: ECG-only, PPG-only, concatenation fusion and the proposed cross-attention fusion. Each signal goes through its own 1D-CNN encoder (weights not shared; channels 32/64/128, kernel sizes 15/9/7, pooling 2, dropout 0.3) that outputs a feature *sequence*. In cross-attention (4 heads, dimension 128, positional encodings) the ECG sequence supplies the queries and the PPG sequence supplies the keys and values. A shallow head with 64 hidden units produces the alarm logit.

Rivals. We added four fusion variants (bidirectional cross-attention, a joint transformer with modality embeddings, bottleneck-token fusion and quality-gated cross-attention). We also added a baseline zoo on a 2-channel early-fusion input: ResNet1D [17], InceptionTime [18], a bidirectional GRU, a TCN [19], a patch transformer and a compact diagonal state-space model. In all, 14 neural architectures were compared.

Classical models. The 266 engineered features cover HRV, ECG morphology proxies, wavelet and spectral features, PPG pulse features, ECG-PPG agreement (such as PTT statistics and beat-count ratios in the last 5 s) and signal-quality indices. They feed six algorithms: logistic regression, RBF-SVM, random forest, extra trees, XGBoost [20] and an MLP. A Pan-Tompkins-style detector finds the ECG beats [21]. For each fold and algorithm, 30 random hyper-parameter configurations are scored on the inner validation set only. We also report a first, 53-feature gradient-boosting baseline ("features v1").

Hybrid deep recipe. This recipe follows the attention CNN-RNN design of [7]; it is our re-implementation from the published description, not the authors' code. A 48-configuration grid (6 architectures $\times$ augmentation $\times$ loss $\times$ learning rate) was declared in advance and scored by inner-validation AUC. The winner was an attention CNN-GRU with augmentation, focal loss and a learning rate of 10^{-3}. We combined it with AdamW [22], a one-cycle schedule [23], weight EMA, test-time augmentation and a 3-member ensemble. We also tested

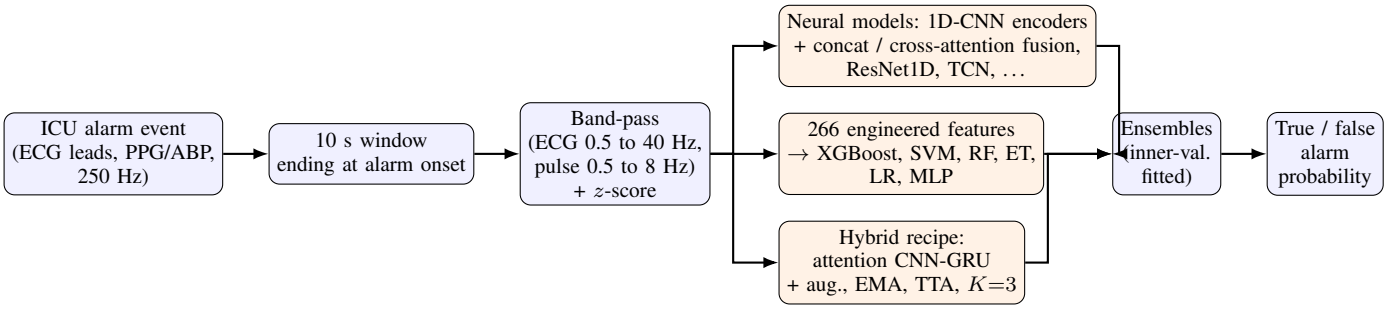


Fig. 1. Overview of the verification pipeline. All stages see only samples up to the alarm onset; hyper-parameters, thresholds and ensemble weights are fitted on inner validation records of each training fold.

TABLE I
DATASETS USED IN THIS STUDY, WITH REQUIRED CITATIONS AND SOURCE URLS.

Dataset	Role	Content used	Citation and source URL
PhysioNet/CinC Challenge 2015 training set	Primary training and cross-validation	750 ICU records, 5 alarm types, expert true/false label; 592-record PPG cohort, 724-record recovered cohort	[6], [10]; https://physionet.org/content/challenge-2015/1.0.0/
VTaC v1.1	Second training/test set, official patient-disjoint split	5,037 annotated VT alarms (3 US hospitals, 3 monitor vendors); 4,542 used	[3], [11]; https://physionet.org/content/vtac/
MIMIC PERform AF	External AF-screening task (35 subjects)	19 AF, 16 non-AF subjects; ECG + PPG at 125 Hz	[12], [13]; https://zenodo.org/records/15906524
MIT-BIH Noise Stress Test	Synthetic noise injection (robustness)	Baseline wander, electrode motion, muscle artifact	[14]; https://physionet.org/content/nstdb/1.0.0/
MIT-BIH Arrhythmia Database	Validation of the R-peak detector	10 records	[15]; https://physionet.org/content/mitdb/1.0.0/
PaPaGei pretrained weights	Foundation-model comparison (not a dataset)	PPG encoders PaPaGei-S/-P	[16]; https://zenodo.org/records/13983110

hybrid variants that append the 53 agreement features to the deep representation.

Ensembles. We tried a logit mean, a probability mean, a logistic-regression stacker and greedy selection. Each is fitted on inner-validation predictions only.

Self-supervised pretraining and foundation models. We pretrained contrastive, masked-reconstruction and cross-modal-prediction objectives on VTaC pre-alarm segments from training and validation patients only. The PaPaGei PPG foundation model [16] was evaluated frozen (linear probe), as a frozen encoder inside fusion models, and with last-block fine-tuning. We downloaded the ECG-FM weights [24] but did not run them. The model needs a 12-lead 500 Hz input and a compiler toolchain that we did not have.

E. Training procedure

Baseline-ladder models use Adam with a learning rate of 10^{-3}, weight decay 10^{-4}, batch size 16, at most 60 epochs and focal loss. Early stopping (patience 10) watches inner-validation F1. The hybrid recipe uses AdamW (weight decay 10^{-2}), a one-cycle schedule with 15% warm-up, gradient clipping at 1.0, batch size 32 (64 on VTaC) and at most 30 epochs (20 on VTaC). It stops early on inner-validation AUC. Training-only augmentation consists of a joint circular time shift of ± 1 s, per-channel amplitude scaling, additive Gaussian noise and a random 1 s dropout of one modality. The ladder's hyper-parameters sit in a configuration file and are not tuned per variant. We handle class imbalance with class weighting or focal loss, not resampling.

F. Implementation details and tools

The code is Python, built on PyTorch [25], scikit-learn [26], XGBoost [20], SciPy, PyWavelets, pandas and the WFDB package for reading PhysioNet records [27]. Experiments ran on a single workstation GPU (RTX PRO 4000 Blackwell, 24 GB) with 20 CPU cores, and neural folds train in seconds to minutes. The repository holds the code, configurations and the result files behind every table and figure.

G. Mathematical formulation

Let $E, P \in \mathbb{R}^{T' \times d}$ be the ECG and PPG feature sequences (with positional encodings added). Cross-attention computes

$$\mathrm{Attn}(E, P) = \mathrm{softmax}\left(\frac{(EW_Q)(PW_K)^\top}{\sqrt{d_k}}\right) PW_V, \quad (1)$$

so each ECG time step collects the PPG steps it finds relevant. Concatenation fusion instead stacks the pooled features. Training uses the focal loss [28]

$$\mathcal{L} = -\alpha_t (1 - p_t)^\gamma \log p_t, \quad \gamma = 2, \quad (2)$$

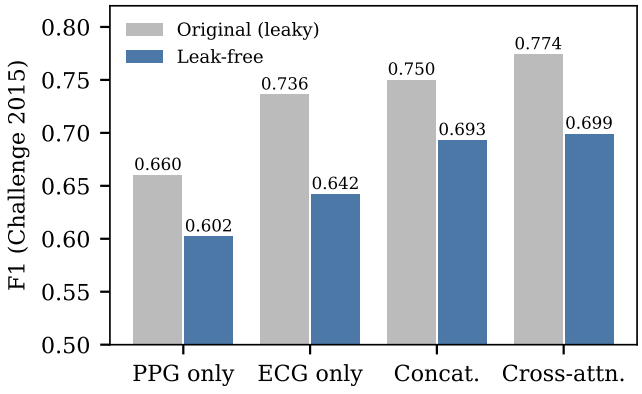


Fig. 2. F1 on Challenge 2015 before and after removing the validation leak.

where p_t is the predicted probability of the true class. The pulse transit time of beat i is $\mathrm{PTT}_i = t_i^{\mathrm{foot}} - t_i^{R}$, the delay from the R-peak to the PPG foot. We score classification with $\mathrm{F1} = 2\,\mathrm{TP}/(2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$, AUC, and the Challenge 2015 score

$$S = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+5\,\mathrm{FN}}, \tag{3}$$

in which a missed true alarm costs as much as five false alarms [6]. For two models evaluated on J paired folds, the Nadeau-Bengio corrected resampled t-statistic [29] is

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}}\right) s_d^2}}, \tag{4}$$

with $\bar{d}$ and s_d^2 the mean and variance of the paired F1 differences. We add a paired record-level bootstrap with 5,000 resamples [30], which applies the same resample to both models, and Holm correction [31] over a comparison family fixed in the configuration file before the final runs. A finding counts as robust only if both tests agree after correction. The safety layer uses temperature scaling [32] and certifies a suppression threshold with an exact Clopper-Pearson upper bound on the miss rate [33]. The repository also holds a TRIPOD+AI reporting checklist [34].

III. Results and Discussion

A. Effect of removing the validation leak

The original training script chose the best epoch, and stopped training, using the same fold it then reported. With a separate inner validation set and 10 seeds, every score fell (Fig. 2). The cross-attention F1 dropped from 0.774 to 0.699, and its edge over concatenation shrank from $+0.024$ to $+0.006$ (bootstrap 95% CI -0.015 to $+0.028$). The leak came from our earlier code, not from the method. We archived the original numbers and use them nowhere else.

B. Leaderboards

Tables II and III and Fig. 3 give the main comparison (threshold 0.5; means over seeds and folds).

TABLE II
Challenge 2015, PPG cohort (592 records), 5-fold CV × 10 seeds.

Model	Acc.	F1	AUC
Ensemble, 10 models (logit mean)	0.876	0.827	0.948
XGBoost, features v2	0.871	0.818	0.939
Extra Trees, features v2	0.867	0.802	0.947
Random Forest, features v2	0.863	0.795	0.938
MLP, features v2	0.857	0.802	0.927
SVM (RBF), features v2	0.852	0.791	0.928
Logistic reg., features v2	0.852	0.801	0.928
HistGB, features v1	0.851	0.798	0.918
ResNet1D	0.820	0.760	0.884
TCN	0.810	0.751	0.887
InceptionTime	0.796	0.732	0.876
CNN cross-attention (proposed)	0.765	0.699	0.841
CNN concatenation	0.757	0.693	0.836
Hybrid deep recipe (10 seeds)	0.871	0.828	0.932
Ensemble incl. hybrid (5 seeds)	0.878	0.830	0.950

TABLE III
VTaC, official split (427 test events), 5 seeds.

Model	Acc.	F1	AUC
Hybrid deep recipe	0.913	0.858	0.966
Ensemble incl. hybrid (logit mean)	0.912	0.844	0.957
Ensemble, 10 models (logit mean)	0.910	0.839	0.955
MLP, features v2	0.900	0.821	0.949
XGBoost, features v2	0.895	0.814	0.942
SVM (RBF), features v2	0.891	0.806	0.948
CNN bidirectional cross-attention	0.857	0.775	0.933
TCN	0.856	0.760	0.919
ResNet1D	0.852	0.750	0.915
CNN cross-attention (proposed)	0.844	0.746	0.915
CNN concatenation	0.837	0.744	0.909

On Challenge 2015 the proposed cross-attention network ranks 12th of the 13 models in Table II by F1. Every classical model on engineered features beats it, as do ResNet1D, TCN and InceptionTime. The best leak-free results come from the 10-model ensemble (accuracy 0.876) and the hybrid deep recipe (F1 0.828, AUC 0.932). The ensemble's F1 lead over the best classical model is within noise ($+0.008$, CI -0.007 to $+0.022$), but its lead over cross-attention is $+0.128$ (CI $+0.094$ to $+0.161$). On VTaC the hybrid recipe leads (F1 0.858, AUC 0.966). The top three models are separated by less than the bootstrap noise, while the gap to cross-attention is $+0.11$ F1 (CI $+0.07$ to $+0.15$). Fusion does help on VTaC: ECG-only reaches F1 0.709 and PPG-only 0.566, against 0.744 for concatenation.

Caveat on the hybrid recipe. We selected its configuration on inner-validation scores of Challenge 2015 folds. Those records are test records in other seeds and folds, so a small optimistic bias is possible. We applied the same recipe to VTaC without re-tuning, and a shuffled-label control gave a chance-level AUC of 0.57, so we found no sign of leakage. The ablation in Table IV shows that weight averaging (EMA)

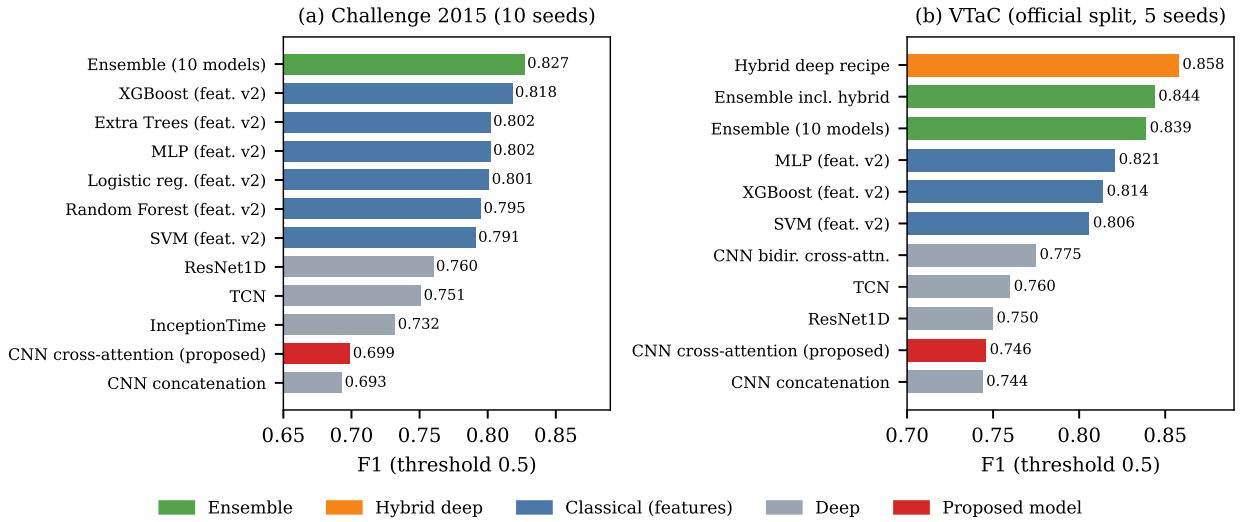


Fig. 3. F1 leaderboards (threshold 0.5). The originally proposed cross-attention model (red) ranks near the bottom on both datasets.

TABLE IV
ABLATION OF THE HYBRID RECIPE, CHALLENGE 2015 (10 SEEDS, PAIRED AGAINST THE FULL RECIPE; SE OVER SEEDS).

Variant	F1	AUC	ΔF1 (SE)
Full recipe (K=3, EMA, TTA)	0.828	0.932	0
No TTA	0.824	0.931	−0.003 (0.002)
No EMA	0.819	0.932	−0.008 (0.002)
Ensemble size 2	0.828	0.930	0.000 (0.002)
Single member (EMA, TTA)	0.815	0.925	−0.012 (0.004)
Single member, raw, no TTA	0.793	0.922	−0.035 (0.010)

TABLE V
PAIRED F1 DIFFERENCES ON CHALLENGE 2015 (CHALLENGER − BASELINE; 50 SEED×FOLD PAIRS). p_{NB}: NADEAU-BENGIO, p_{boot}: BOOTSTRAP; BOTH HOLM-ADJUSTED.

Baseline	Challenger	ΔF1	95% CI	p_{NB}	p_{boot}
Concat.	Cross-attn.	+0.006	[−0.015, +0.028]	1.00	1.00
ECG-only	Cross-attn.	+0.058	[+0.028, +0.088]	1.00	0.003
PPG-only	Cross-attn.	+0.094	[+0.058, +0.129]	0.077	<0.001
Cross-attn.	Bidirectional	−0.008	[−0.023, +0.008]	1.00	1.00
Cross-attn.	Joint transf.	−0.006	[−0.025, +0.013]	1.00	1.00
Cross-attn.	Gated	−0.002	[−0.015, +0.012]	1.00	1.00
Cross-attn.	Bottleneck	−0.047	[−0.070, −0.026]	1.00	<0.001
ECG-only	ResNet1D	+0.118	[+0.089, +0.148]	0.052	<0.001
ResNet1D	Cross-attn.	−0.060	[−0.087, −0.035]	0.95	<0.001
TCN	Cross-attn.	−0.052	[−0.080, −0.023]	1.00	<0.001
GBM (feat. v1)	Cross-attn.	−0.099	[−0.133, −0.066]	0.049	<0.001

is the largest single post-training ingredient. The baselines use a different trainer, so part of the gap to the baseline CNNs reflects the trainer (optimizer, regularization, ensembling) and not the architecture.

C. Pre-specified paired comparisons

Table V and Fig. 4 report the comparison family that we fixed before the final runs.

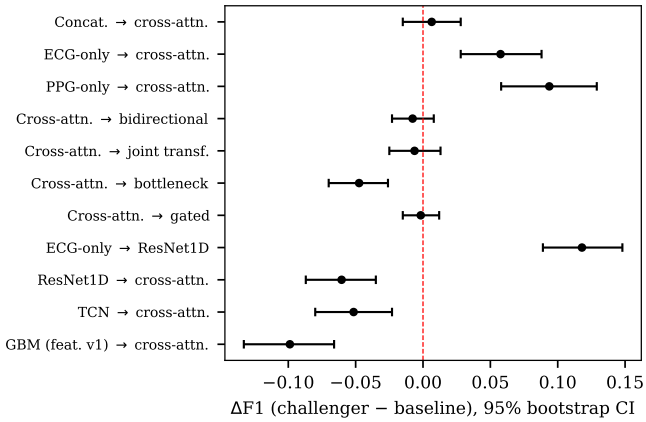


Fig. 4. Paired F1 differences with 95% bootstrap confidence intervals (Challenge 2015).

The four attention-style fusion variants cannot be told apart from each other or from concatenation. The bootstrap supports a benefit of fusion over either single signal, but the more conservative Nadeau-Bengio test does not after Holm correction. Under our rule that both tests must agree, only comparisons such as GBM versus cross-attention count as robust. The two tests differ as expected, because the corrected t-test accounts for overlapping training sets.

D. Per-alarm-type behavior

Table VI shows pooled out-of-fold results for cross-attention. Performance depends heavily on the alarm type. Tachycardia has an F1 of 0.92 but a specificity of only 0.42, and VT has an F1 of 0.49. Asystole and ventricular flutter/fibrillation have too few true alarms (14 and 5) for reliable conclusions. In the error taxonomy, VT dominates the persistent missed true alarms: 18 of 29 persistent false

TABLE VI
CROSS-ATTENTION PER ALARM TYPE (CHALLENGE 2015, POOLED OUT-OF-FOLD).

Alarm type	n	True	F1	Sens.	Spec.
Asystole	93	14	0.41	0.56	0.80
Bradycardia	73	39	0.75	0.69	0.83
Tachycardia	111	106	0.92	0.87	0.42
V. flutter/fib.	45	5	0.26	0.42	0.77
V. tachycardia	270	59	0.49	0.58	0.78

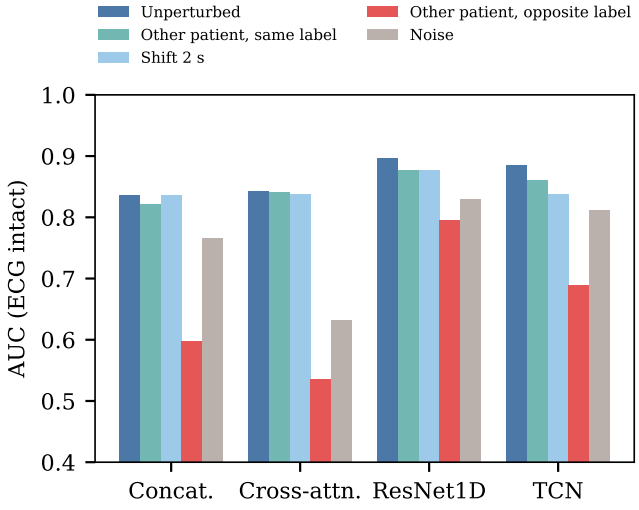


Fig. 5. Test AUC when the PPG is perturbed and the ECG is intact.

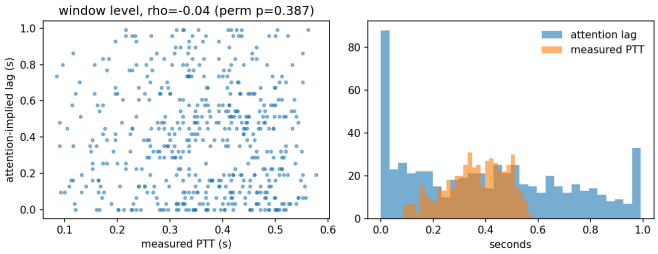


Fig. 6. Attention-implied lag versus measured PTT (left, window level) and the two lag distributions (right).

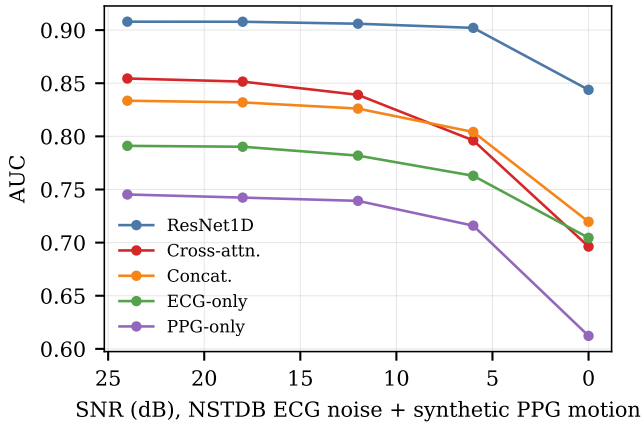


Fig. 7. AUC versus injected noise level (mean over noise types; seed-0 checkpoints).

negatives for the feature GBM, although VT makes up only 26% of true alarms.

E. Do fusion models use ECG-to-pulse timing?

The original design rests on this question. We perturbed one modality while keeping the other intact (seed-0 checkpoints, five test folds; Fig. 5). Replacing the PPG with another patient's *same-label* PPG, whose timing is unrelated to the ECG, changes AUC by only 0.00 to 0.03 (cross-attention 0.002, concatenation 0.015, ResNet1D 0.021, TCN 0.024). Shifting the PPG by up to 2 s changes AUC by at most 0.005 for the CNN fusion models and by 0.03 to 0.06 for ResNet1D, TCN, InceptionTime and BiGRU. An *opposite-label* pulse, by contrast, collapses AUC (cross-attention 0.842 to 0.535). The models use the *content* of each signal (morphology, regularity, presence of a pulse) as evidence, not the beat-to-pulse timing.

The attention maps tell the same story. The attention-implied lag has no relationship with the measured PTT (window-level Spearman $\rho = -0.039$, permutation $p = 0.39$; beat-level $\rho = 0.017$). Only 5.3% of attention mass falls in the physiological lag range, against 5.6% for uniform attention (Fig. 6). In deletion and insertion tests, occlusion is clearly faithful (PPG deletion -0.041, insertion $+0.041$) and integrated gradients modestly so (-0.016/$+0.010$). Attention weights are barely better than a random ordering (-0.007/$+0.011$). They should not be presented as an explanation, and these data do not support the pulse-transit-time rationale in our original proposal.

F. Robustness to noise and sensor loss

We added real Noise Stress Test noise to the ECG and synthetic motion artifact to the PPG (SNR 24 to 0 dB). ResNet1D is the most robust (AUC 0.907 clean to 0.844 at 0 dB on both signals), while cross-attention falls from 0.856 to 0.696 and concatenation from 0.833 to 0.720 (Fig. 7). Zeroing the PPG at inference, without retraining, costs cross-attention 0.12 AUC (0.856 to 0.736), concatenation 0.10 and ResNet1D 0.27. Zeroing the ECG costs cross-attention 0.08. Constant PPG delays of 0 to 400 ms barely change the CNN fusion models, which fits the previous subsection. A check that the injected noise is not a learnable shortcut gave an AUC of 0.52 to 0.55 from model probabilities alone.

G. Generalization across datasets

Transfer is strongly asymmetric (Fig. 8). Models trained on the small Challenge 2015 set and tested on the VTaC test split reach an F1 of 0.56 to 0.60 (AUC 0.73 to 0.78) when they use the ECG. The PPG-only model is near chance (F1 0.432, AUC 0.541), while in-distribution VTaC F1 is 0.71 to 0.78. In the other direction, models trained on VTaC transfer well to Challenge 2015 VT alarms (ResNet1D F1 0.773, AUC 0.927). Cross-attention trained and tested within Challenge

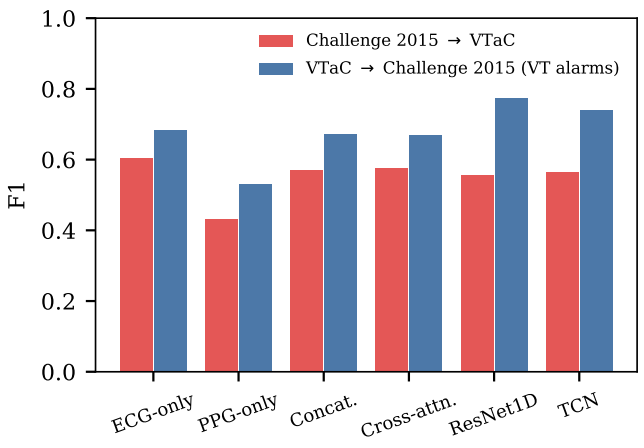


Fig. 8. Cross-dataset transfer (F1).

2015 reaches a VT F1 of only 0.49. The amount of training data therefore matters more than the model family, and single-dataset numbers overstate real-world performance.

H. Pretraining, foundation models and AF screening

Fine-tuning from self-supervised encoders gives modest paired gains over scratch training (+0.01 to +0.04 F1, three seeds, label fractions 10 to 100%). Only 1 of 72 paired contrasts has a Nadeau-Bengio $p < 0.05$, so the effect is suggestive, not established. The PaPaGei foundation model did not beat from-scratch encoders. On Challenge 2015 a frozen linear probe reached F1 0.58 (PPG alone), and the best PaPaGei fusion model sat about 0.04 F1 below scratch cross-attention. On VTaC it improved on a scratch PPG-only model (F1 0.58 to 0.62 versus 0.566) but not on scratch fusion (0.745). On the MIMIC PERform AF task (35 subjects, subject-wise cross-validation, low power), a gradient-boosting model on RR-irregularity features reached subject-level AUC 1.00 and was significantly better than the best neural model (0.93). Fusion beat a PPG-only network by $+0.28$ subject AUC. Distilling a PPG-only student from the fusion teacher gave no benefit over hard labels.

I. Label reliability, time-to-verdict and safety

On VTaC, 7.7% of events (351 of 4,542) have split annotator votes, and accuracy on them is much lower (cross-attention 0.704 versus 0.854 on unanimous events). Model entropy tracks disagreement only weakly (Spearman 0.13 to 0.18). Training on annotator vote fractions improved cross-attention by $+0.038$ F1 (SD 0.012, 5 seeds). The time-to-verdict analysis slides the 10 s window past the trigger, and F1 falls (cross-attention 0.740, 0.713, 0.515 and 0.451 at 0, 5, 10 and 30 s). The triggering event leaves the window as it slides, so this does not show whether waiting helps. We did not run a design that appends post-alarm seconds to the pre-alarm window.

We ran a safety layer (temperature scaling, a risk-controlled threshold with a Clopper-Pearson bound, and a deferral band) for the concatenation model only. A 10% miss tolerance at 90% confidence could be certified in 42 of 50 folds. In those folds the layer suppressed 22.5% of false alarms while missing 2.9% of true alarms, and calibration reduced the expected calibration error from 0.167 to 0.121. Thresholds aimed at 95 to 100% validation sensitivity realized only about 92 to 97% on test folds, so sensitivity targets must not be assumed to transfer.

J. Limitations

(i) The data are small. Fold-to-fold F1 standard deviation is about 0.05, many differences between top models sit inside noise, and the standard deviations over seeds$\times$folds are not confidence intervals. (ii) Several engineered features encode the alarm definitions, and so does the alarm type itself on Challenge 2015. That is built-in domain knowledge, not learned physiology, and we did not run an ablation without the alarm type. (iii) The hybrid recipe carries a small selection-bias risk. (iv) The Mousavi-style and contrastive models are our re-implementations from the published descriptions. Only a two-seed sanity run of the former exists, so we make no claim about the original methods' performance. (v) There is no prospective validation, and all data come from US ICUs. We did not assess the effects of demographics and skin tone on optical PPG. VTaC device identifiers are not released (the lead-set signature we used as a proxy is unverified), and patient overlap between corpora cannot be checked. (vi) Because of time or data access, we did not run leave-one-device-out evaluation, long pre-alarm context, ECG-FM, or pretraining on larger corpora (PulseDB, VitalDB, MIMIC-III). (vii) The model must not be used to suppress alarms unsupervised. The data are public and de-identified under the sources' IRB approvals.

IV. Conclusion

We tested whether cross-attention fusion of ECG and pulse signals verifies ICU arrhythmia alarms better than simpler alternatives. Under a leakage-free protocol it does not. The originally proposed model is no better than naive concatenation. Engineered-feature classifiers, a carefully trained hybrid attention CNN-GRU and ensembles clearly beat it (accuracy 0.878 to 0.913, F1 0.83 to 0.86). The two signals do complement each other, since fusion beats either signal alone on VTaC. The models, however, use the content of each signal and not the beat-to-pulse timing that motivated the design, and the counterfactual tests and the attention-PTT analysis agree on this. An unnoticed validation leak had inflated our headline F1 by 0.075, and the controlled protocol is what makes the remaining conclusions trustworthy. Next steps are leave-one-site-out evaluation, larger unlabeled pretraining corpora and prospective validation. The code, configurations and result files are in the project repository.

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